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**PEDIATRIC VS ADULT DENGUE SEVERITY
PREDICTION WITH SUBGROUP-SPECIFIC
SHAP EXPLANATIONS**

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DECLARATION

This is certifying that the thesis titled “Pediatric vs Adult Dengue Severity Prediction with Subgroup-Specific SHAP Explanations” is our own research carried out under the guidance of our supervisor Mohammad Arif Hasan Chowdhury, Assistant Professor, Department of Computer Science and Engineering, University of Science and Technology Chittagong. This thesis has not been submitted, either in whole or in part, for the award of any degree or diploma at any other university or institution.

All sources of information and materials used in this study have been properly acknowledged through appropriate references. Any assistance received in the course of this research has been duly acknowledged. The results, analysis, and conclusions presented in this thesis are original and based on the experimental work conducted by us.

We further declare that no part of this thesis has been copied from any other source without proper citation and that the work complies with the academic integrity and ethical standards of the university.

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APPROVAL CERTIFICATE

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DEDICATION

This project is dedicated to all the parents.

ABSTRACT

Dengue fever progression to severe illness poses an immense global public health burden. Machine learning models developed for dengue severity risk assessment frequently pool pediatric and adult patients into a single uniform cohort, assuming that feature-severity relationships remain invariant across age demographics. This thesis investigates whether partitioning clinical datasets into pediatric (≤ 18 years) and adult (> 18 years) cohorts alters classification performance and SHapley Additive exPlanations (SHAP) feature attribution patterns compared to an age-pooled baseline.

Because the supplied datasets lacked adjudicated WHO severity outcomes and clinical warning-sign variables, a Niazi–Momand hematology-based proxy was constructed from platelet count, hematocrit, and WBC thresholds. Platelet and all platelet-derived predictors were strictly excluded from the model feature matrix to reduce circular target leakage. The resulting performance estimates describe proxy classification and should not be interpreted as clinical diagnosis or WHO severe-dengue prediction.

Using a harmonized clinical dataset of 2,032 laboratory-confirmed Dengue NS1-positive patient records, five supervised classifiers (Logistic Regression, Random Forest, XGBoost, LightGBM, and Stacking Ensemble) were benchmarked across Pooled, Pediatric, and Adult cohorts using Repeated Stratified Cross-Validation. Random Forest and Stacking Ensembles demonstrated strong stability. Extracting decision tree splits on positive SHAP contributions highlighted age-divergent biomarker thresholds, confirming that models trained on an age-pooled proxy often obscure distinct age-conditioned physiological boundaries.

To address generalizability and prevent duplicated row inflation, stringent dataset provenance and deduplication hashing protocols were implemented. Finally, to bridge analytical findings with research prototype capabilities, the leakage-controlled models were integrated into a full-stack clinical decision support web application developed with React 19 and FastAPI, featuring real-time SHAP waterfall attributions.

Keywords: Age Stratification, Hematology Severity Proxy, Explainable AI, Zero Leakage, Machine Learning, Clinical Decision Support System.

Contents

DECLARATION	i
APPROVAL CERTIFICATE	ii
ACKNOWLEDGEMENT	iv
DEDICATION	v
ABSTRACT	vi
1 INTRODUCTION	1
1.1 Background and Motivation	1
1.2 Problem Statement	2
1.3 Research Objectives	2
1.4 Objectives of the Study	3
1.4.1 Main Objective	3
1.4.2 Specific Objectives	3
1.5 Research Questions	4
1.6 Scope and Boundaries	4
1.7 Summary of Contributions	4
2 LITERATURE REVIEW	6
2.1 Overview of Automated Dengue Severity Prognosis	6
2.2 Classical Machine Learning & Ensemble Architectures	6
2.3 Explainable Artificial Intelligence (XAI) in Dengue	7
2.4 Longitudinal Tracking and Temporal Snapshot Analysis	7
2.5 Demographic Subgrouping and Generalizability	7
2.6 Challenges with Imbalanced Clinical Datasets	8
2.7 Feature Selection and Dimensionality Reduction	8
2.8 Full-Stack Integration and Clinical Decision Support Systems	8
2.9 Literature Synthesis and Comparative Matrix	9
2.10 Identified Research Gaps	10

3	METHODOLOGY	12
3.1	Overall Research Architecture	12
3.2	Dataset Description and Cohort Partitioning	12
3.3	Data Preprocessing and Quality Control	14
3.3.1	1. Positive Dengue NS1 Filtering	14
3.3.2	2. Cryptic Duplicate Removal	14
3.3.3	3. Missing Value Imputation	14
3.4	Target Harmonization & Zero-Leakage Mathematical Guarantee	15
3.5	Feature Engineering	15
3.6	Classification Algorithms and Formulations	16
3.6.1	Logistic Regression (LR)	16
3.6.2	Random Forest (RF)	16
3.6.3	XGBoost (XGB)	16
3.6.4	LightGBM (LGBM)	16
3.6.5	Stacking Ensemble Classifier	17
3.7	Explainable AI and Programmatic Threshold Induction	17
3.7.1	Shapley Value Formulation	17
3.7.2	Depth-1 Decision Tree Threshold Extraction	17
3.7.3	Rank Correlation Analysis	17
3.8	System Development and Implementation	18
3.8.1	System Architecture Overview	18
3.8.2	Frontend Web Application Development (React 19)	18
3.8.3	Backend RESTful API Engine (FastAPI)	18
3.8.4	Database Schema and Longitudinal Storage (SQLite)	20
3.8.5	Cloud Infrastructure and Security	20
4	RESULTS AND ANALYSIS	22
4.1	Exploratory Data Analysis and Cohort Divergence	22
4.2	Model Performance Evaluation	22
4.3	Performance Curve Verification	24
4.4	Target Leakage Baseline Comparison	24
4.5	SHAP Global Feature Hierarchies	26
4.6	Rank Correlation and Model-Derived Decision Splits	28
4.7	Independent External Dataset Validation	30
5	DISCUSSION	31
5.1	Synthesis of Research Outcomes	31
5.2	Biological & Pathophysiological Interpretation	31
5.3	External Validation and Cross-Dataset Robustness	32

5.4	Clinical Decision Support System (CDSS) Dashboard Implementation . .	32
5.5	Limitations and Methodological Boundaries	32
5.5.1	1. Retrospective Platelet-Derived Severity Proxy vs. Clinical WHO Gold Standard	32
5.5.2	2. Geographic, Demographic, and Viral Serotype Spectrum Bias .	34
5.5.3	3. Pediatric Cohort Sample Size and Cross-Validation Granularity	36
5.5.4	4. Cross-Sectional Laboratory Snapshots vs. Continuous Temporal Dynamics	36
5.5.5	5. Clinical Usability Validation and Interface Boundaries	36
6	CONCLUSION AND FUTURE WORK	37
6.1	Conclusion	37
6.2	Summary of Key Contributions	38
6.3	Future Work and Recommendations	38

List of Figures

3.1	End-to-end research architecture: Data ingestion, NS1 filtering, zero-leakage feature engineering, age-stratified model benchmarking, SHAP threshold extraction, and full-stack deployment.	13
3.2	Comprehensive Clinical Decision Support System architecture: Interactive input form, dynamic risk gauge, local SHAP waterfall visualizer, and multi-assessment temporal snapshot tracking.	19
3.3	Entity-Relationship Schema for the SQLite Longitudinal Storage.	21
4.1	Exploratory data analysis comparing distributions of Platelet count, WBC count, Hematocrit, and Lymphocyte percentage between Pediatric and Adult cohorts.	23
4.2	Learning Curve for the Zero-Leakage Proxy Model (Random Forest), demonstrating stable convergence between training and cross-validation Macro-F1 scores without significant overfitting as sample size increases.	25
4.3	Confusion Matrix heatmaps for best-performing Random Forest models on held-out Pediatric and Adult test sets, highlighting zero false positives in pediatrics.	25
4.4	Target Leakage Artifact vs Genuine Physiological Proxy. The red curve demonstrates artificially inflated performance when including target-derived variables (Platelets), while the blue curve represents the true zero-leakage predictive capability.	26
4.5	SHAP summary beeswarm plots comparing feature attributions across Pooled, Pediatric, and Adult cohorts.	27
4.6	SHAP dependence plots for WBC count in Pediatric (left) and Adult (right) cohorts.	29
4.7	SHAP dependence plots for AST in Pediatric (left) and Adult (right) cohorts, showing sharp pediatric threshold surge at ~ 50 U/L.	29
4.8	Independent external dataset validation ROC curve ($N = 50$), demonstrating high generalizability ($AUC = 0.980$, $Accuracy = 96.0\%$).	30

5.1	Interactive Patient Assessment Form with real-time biomarker validation and automated age cohort routing.	33
5.2	Longitudinal Patient History view showing day-to-day snapshot comparisons and delta change tracking.	34
5.3	Patient-Level Local SHAP Waterfall plot showing individual feature risk pushes toward severe vs non-severe prognosis.	35
5.4	Pediatric vs. Adult Cohort Comparison mode evaluating identical patient parameters across age-specific models.	35

List of Tables

2.1	Comprehensive Synthesis of Dengue Severity Machine Learning Literature	9
4.1	Classification Performance Benchmark Across Cohorts (Repeated Stratified 5-Fold CV)	24
4.2	Top Clinical Features Ranked by Mean Absolute SHAP Attribution (mean SHAP) Across Cohorts	28

Chapter 1

INTRODUCTION

1.1 Background and Motivation

Dengue fever is an acute mosquito-borne viral infection caused by four antigenically distinct serotypes (DENV 1–4) of the genus *Flavivirus* and transmitted predominantly by *Aedes aegypti* and *Aedes albopictus* vectors. Over the past two decades, the global incidence of dengue has grown dramatically, with the World Health Organization (WHO) estimating that more than 3.9 billion people across 129 countries are at risk of infection annually [1, 2]. In South Asian endemic regions, particularly Bangladesh, massive seasonal outbreaks characterized by hyperendemic multi-serotype circulation have placed severe pressure on tertiary hospital infrastructures.

The clinical trajectory of dengue infection spans a broad spectrum, ranging from self-limiting febrile illness to severe dengue manifestations. Severe dengue is characterized pathophysiologically by sudden plasma leakage resulting from endothelial hyperpermeability, severe thrombocytopenia, massive internal hemorrhage, and multiorgan impairment including acute hepatic failure and myocarditis [1]. Because the critical phase typically begins around the time of defervescence (days 3–7 of illness), early identification of patients likely to progress to life-threatening complications is vital for timely volume replacement and fluid resuscitation.

Supervised machine learning algorithms have emerged as powerful clinical decision support tools capable of synthesizing routine clinical hematological and biochemical markers to predict severe outcomes [15, 16, 8]. However, existing prognostic models universally suffer from a fundamental architectural limitation: they pool pediatric and adult patients into a single training cohort, treating patient age merely as an ordinary continuous input variable.

Clinical literature in tropical medicine demonstrates that dengue pathophysiology diverges fundamentally between pediatric and adult populations [4, 3, 5]:

1. **Pediatric Immune Reactivity:** Children exhibit enhanced baseline microvas-

cular fragility, immature immune regulation, and distinct lymphoid distributions, resulting in rapid fluid loss, acute reactive lymphocytosis, and sudden progression to Dengue Shock Syndrome (DSS).

2. **Adult Comorbidity & Hepatic Involvement:** Adults possess mature vascular baselines but frequently suffer from secondary comorbidities, prominent hepatic transaminase elevations ($AST > ALT$), and marked hemoconcentration (HCT) driven by vascular plasma extravasation.

Collapsing these two distinct physiological populations into an age-pooled machine learning framework forces models to learn compromised global boundaries that obscure age-specific predictive signals. Consequently, there is an urgent need for an age-stratified, explainable machine learning methodology.

1.2 Problem Statement

Despite extensive machine learning research in infectious diseases, existing dengue severity prediction frameworks exhibit four major limitations:

1. **Unavailable Clinical WHO Labels in Public Data:** Many public datasets lack clinically adjudicated severity labels (warning signs, plasma leakage, shock), forcing models to either discard data or mistakenly label patients.
2. **Circular Target Leakage in Prior Literature:** Numerous published studies use laboratory variables (e.g., platelet count) to define severe dengue proxy labels, while simultaneously using those same variables as predictive inputs. This circular leakage heavily inflates reported performance.
3. **Age-Pooled Model Homogenization:** Current models train unified classifiers across mixed-age cohorts, assuming that hematological biomarker distributions and risk thresholds operate identically across infants, adolescents, and older adults.
4. **Global-Only Feature Attributions:** Standard explainable AI (XAI) implementations calculate global SHAP summaries across pooled cohorts, concealing subtle subgroup-conditioned feature importance shifts.

1.3 Research Objectives

This thesis aims to study differences in prediction of a hematology-derived proxy across age subgroups. The specific objectives are:

1. **Data Harmonization:** Harmonize and deduplicate multiple public dengue datasets utilizing cryptographic hashing to freeze data provenance.
2. **Proxy Target Construction:** Construct a reproducible Niazi–Momand hematology proxy target and explicitly compare it with WHO clinical severity requirements.
3. **Zero-Leakage Enforcement:** Prevent circular data leakage by completely excluding platelet counts and platelet-derived features from the predictive feature matrix.
4. **Age-Stratified Benchmarking:** Compare pooled and age-stratified models with robust repeated cross-validation.
5. **Explainability (XAI):** Quantify SHAP attribution stability to understand which non-platelet hematology variables contribute most to proxy classification within each age cohort.

1.4 Objectives of the Study

1.4.1 Main Objective

To design, implement, and validate an age-stratified explainable machine learning framework for early-stage dengue severity prediction that prevents target leakage, quantifies subgroup-specific SHAP feature attributions, and provides a full-stack clinical decision support dashboard.

1.4.2 Specific Objectives

- **SO1 (Data & Feature Pipeline):** Construct a zero-leakage preprocessing and feature engineering pipeline on $N = 972$ laboratory-confirmed Dengue NS1 patients from Bangladesh, completely isolating platelet counts from predictive inputs.
- **SO2 (Age-Stratified Benchmarking):** Train and benchmark five machine learning classifiers (Logistic Regression, Random Forest, XGBoost, LightGBM, Stacking Ensemble) across Pediatric (≤ 18 years), Adult (> 18 years), and Pooled cohorts.
- **SO3 (Explainable AI & Threshold Extraction):** Compute localized TreeExplainer SHAP attributions, calculate Kendall’s τ -b rank correlation, and extract explicit age-divergent non-linear decision tree split thresholds for key hematological biomarkers.
- **SO4 (External Validation):** Evaluate model transferability and generalizability on independent external cohorts to establish cross-dataset robustness.

- **SO5 (Full-Stack Clinical Deployment):** Develop and deploy a full-stack web application using React 19 and FastAPI featuring real-time risk scores, interactive SHAP waterfalls, and longitudinal patient tracking.

1.5 Research Questions

- **RQ1:** Does age-stratified cohort partitioning improve predictive classification metrics (AUC, Accuracy, F1, Specificity) compared to age-pooled modeling?
- **RQ2:** How do global SHAP feature importance hierarchies differ between pediatric and adult dengue cohorts?
- **RQ3:** Can programmatic single-depth decision tree fits on SHAP values extract actionable, divergent clinical cutoff thresholds between age groups?
- **RQ4:** How robustly do the trained models generalize when tested on independent external clinical cohorts?

1.6 Scope and Boundaries

This study evaluates supervised machine learning models on laboratory-confirmed Dengue NS1-positive patients ($N = 2032$) from Bangladesh. Clinical severity is modeled using the established Niazi & Momand proxy criterion, which defines severe dengue as severe thrombocytopenia ($PLT < 50.0 \times 10^3/\mu L$) corroborated by concurrent hematocrit elevation ($HCT > 45\%$) or extreme leukogram abnormalities ($WBC < 2$ or > 20). The models utilize complete blood count (CBC) and liver transaminase parameters available at presentation. The clinical decision support system is designed as an investigative research prototype for risk triaging and structured longitudinal record collection, and is not intended to replace formal medical diagnosis by a licensed physician.

1.7 Summary of Contributions

The primary contributions of this thesis are:

1. **Age-Stratified Prognostic Framework:** Proving that age-stratified modeling achieves superior predictive accuracy (90.6% Accuracy, 0.947–0.961 AUC in pediatrics) compared to pooled baselines (0.860 AUC).
2. **Zero Target Leakage Benchmark:** Establishing a rigorous leakage-free feature protocol that isolates platelet counts from the feature matrix X .

3. **Programmatic SHAP Threshold Extraction:** Discovering age-divergent physiological split points ($> 44.5\%$ pediatric lymphocyte split vs $> 25.0\%$ adult; ~ 50 U/L pediatric AST surge vs ~ 80 U/L adult).
4. **Independent External Validation:** Validating cross-dataset generalizability on external patient cohorts (96.0% accuracy, 0.980 AUC).
5. **Production Full-Stack Deployment:** Delivering a deployed React 19 + FastAPI clinical dashboard with live SHAP waterfall visualizations.

Chapter 2

LITERATURE REVIEW

2.1 Overview of Automated Dengue Severity Prognosis

Automated machine learning approaches for dengue diagnosis and severity stratification have progressed through three distinct technological eras over the past two decades:

1. **Rule-Based & Statistical Classifiers (2000–2015):** Early research focused on decision tree induction and linear discriminant analysis to establish basic diagnostic heuristics based on white blood cell count and hematocrit [3, 4].
2. **Ensemble & Deep Learning Pipelines (2016–2022):** Adoption of Random Forests, XGBoost, and Artificial Neural Networks to maximize raw classification accuracy on hospital registries [14, 15].
3. **Explainable AI (XAI) & Biomarker Discovery (2023–2026):** Integration of game-theoretic SHAP and LIME algorithms to provide clinician-interpretable feature attributions [16, 17, 8].

2.2 Classical Machine Learning & Ensemble Architectures

Tanner et al. [3] pioneered decision-tree algorithms on early-phase laboratory indicators in a Singapore pediatric cohort ($n = 1,200$), demonstrating that simple tree rules could identify severe cases with reasonable sensitivity. Potts et al. [4] applied logistic regression with asymmetric misclassification cost ratios (1:10) to prioritize severe pediatric dengue in Thailand, finding that platelet count and elevated AST were primary determinants.

More recently, ensemble architectures have dominated the literature. Huang et al. [14] benchmarked ANN, Random Forest, and Gradient Boosting on a Taiwanese cohort

($n = 798$), reporting an ANN AUC of 0.83 using demographic and laboratory variables. Stacking ensemble frameworks developed by Masum et al. [16] and Tuhin et al. [17] on Bangladeshi patient data achieved over 83% and 96% accuracy respectively by combining multiple heterogeneous base learners. Arrubla-Hoyos et al. [21] evaluated stacked ensembles on a large Colombian national registry ($n = 53,814$), achieving 88% accuracy for WHO 2009 severity classification.

2.3 Explainable Artificial Intelligence (XAI) in Dengue

While high predictive accuracy is necessary, clinical adoption requires interpretability. Chowdhury et al. [15] applied XGBoost with SHAP dependence analysis to analyze global factor contributions across a multi-country dataset. Niazi & Momand [8] evaluated hematology-driven severity models across 1,450 public records using SHAP summary plots. Haque et al. [19] proposed a multi-perspective framework combining SHAP, LIME, and Morris sensitivity analysis, while Joly et al. [20] utilized SHAP to explore phenotypic disease associations.

2.4 Longitudinal Tracking and Temporal Snapshot Analysis

While static machine learning classifiers achieve high accuracy using a single cross-sectional admission snapshot, dengue fever is an acutely dynamic disease that evolves rapidly over a 7–10 day clinical course. A single time-point assessment often fails to capture the velocity of hematological deterioration, such as sudden platelet drops or surging hematocrit levels indicative of imminent plasma leakage. Recent research has underscored the necessity of temporal progression modeling. For instance, Madewell et al. [23] utilized longitudinal data snapshots to track day-to-day transitions in clinical status in Puerto Rico. Despite these advancements, full-stack systems rarely implement longitudinal tracking dashboards that allow clinicians to visualize temporal delta changes and trajectory shifts across sequential data snapshots. Consequently, operationalizing temporal snapshots in clinical interfaces remains a critical gap in automated dengue decision-support systems.

2.5 Demographic Subgrouping and Generalizability

Despite the proliferation of XAI frameworks, virtually all existing studies train a single model on an age-pooled cohort. Rajapakse et al. [5] performed unsupervised clustering on 572 patients, identifying age as the single most discriminative variable separating clinical

phenotypes. Sangkaew et al. [22] designed individualized pediatric risk calibration during the early febrile phase. Furthermore, Lu et al. [18] evaluated cross-dataset transferability across multiple international dengue registries, showing severe performance degradation (AUC drops from 0.89 to 0.55) when models trained on one population were naively applied to another.

2.6 Challenges with Imbalanced Clinical Datasets

Dengue severity datasets are inherently imbalanced, as severe cases typically comprise only 5–15% of total hospital admissions. Training machine learning models on unadjusted clinical data often produces classifiers heavily biased toward the majority class (non-severe). To mitigate this, researchers have employed various data augmentation techniques. For instance, Masum et al. [16] applied Synthetic Minority Over-sampling Technique (SMOTE) to balance classes before training ensemble models. However, synthetic generation of hematological data introduces risks of distorting the underlying physiological distributions, potentially generating unrealistic patient profiles that do not exist in real-world triage scenarios. Consequently, advanced algorithmic adjustments, such as cost-sensitive learning [4] and stratified cross-validation, have emerged as preferable alternatives to pure synthetic oversampling.

2.7 Feature Selection and Dimensionality Reduction

The complexity of routine blood counts and biochemical panels often introduces multicollinearity and noise, degrading model performance. Effective feature selection is critical for identifying the most discriminative biomarkers while minimizing computational overhead. Early rule-based studies [3] relied on statistical correlation matrices to trim redundant variables. While dimensionality reduction techniques like Principal Component Analysis (PCA) successfully compress data, they obfuscate the original clinical variables, rendering the resulting features uninterpretable to physicians. Therefore, domain-driven feature engineering combined with tree-based intrinsic feature selection is heavily favored in modern clinical AI to preserve interpretability.

2.8 Full-Stack Integration and Clinical Decision Support Systems

While substantial research has focused on algorithmic optimization and theoretical accuracy, there remains a massive translational gap between trained machine learning models and operational clinical utility. The majority of published dengue severity models exist

solely as offline scripts, inaccessible to frontline healthcare workers. A robust Clinical Decision Support System (CDSS) requires translating predictive algorithms into scalable, low-latency web architectures. Recent conceptual frameworks [19] advocate for real-time prediction capabilities, but very few studies implement end-to-end full-stack solutions. Bridging this gap necessitates the deployment of RESTful APIs and interactive frontend dashboards that can process patient data instantly, visualize SHAP attributions in real-time, and seamlessly integrate into existing hospital triage workflows.

2.9 Literature Synthesis and Comparative Matrix

Table 2.1 provides a systematic comparison of representative studies in dengue machine learning, highlighting the critical research gap regarding age-stratified XAI modeling.

Table 2.1: Comprehensive Synthesis of Dengue Severity Machine Learning Literature

Study	Cohort / Sample	Methods	Outcome Definition	Treatment of Age / Limitations
Tanner et al. [3]	Singapore, $n = 1200$	Decision Trees	Early diagnosis/outcome	Pediatric-only; pre-SHAP era.
Potts et al. [4]	Thailand, $n = 244$	Cost-sensitive LR	Pediatric severe dengue	Pediatric-only; no adult comparison.
Rajapakse et al. [5]	Brazil, $n = 572$	K-means clustering	Clinical phenotypes	Age identified as top cluster feature; no ML prediction.
Huang et al. [14]	Taiwan, $n = 798$	ANN, RF, GBM + SHAP	Severe dengue (binary)	Age-pooled; global SHAP summary only.
Chowdhury et al. [15]	Multi-country, $n = 1858$	XGBoost + SHAP	WHO severity grades	Age-pooled; single global feature importance.
Masum et al. [16]	Bangladesh, $n = 600$	Stacking + SMOTE + SHAP	Severe dengue	Age-pooled; SMOTE distortion risk.

Study	Cohort / Sample	Methods	Outcome Definition	Treatment of Age / Limitations
Tuhin et al. [17]	Bangladesh, $n = 1523$	Stacking + SHAP/LIME	Multi-class severity	Age-pooled; no subgroup-conditioned cutoffs.
Niazi & Mo- mand [8]	Public, $n = 1450$	LR, RF, XGB + SHAP	Combined hematology proxy (PLT, HCT, WBC)	Age-pooled; global SHAP only.
Arrubla-Hoyos et al. [21]	Colombia, $n = 1200$	XGBoost + SVM	Severe vs Non-severe	Uses WHO endpoints, adult biased.
Madewell et al. [23]	Puerto Rico, $n = 354$	Longitudinal ML	Severe dengue transitions	Temporal focus; lacks demographic stratification.
Lu et al. [18]	Multi-dataset	Logistic Regression	Cross-dataset transfer	Demonstrates severe cross-cohort fragility.
Sangkaew et al. [22]	Vietnam/Thai, $n = 2040$	Calibrated ML	Febrile-phase progression	Pediatric-only calibration; no adult baseline.
Our Study	Bangladesh, $N = 2032$	Age-Stratified Ensembles + SHAP Splits	Zero-Leakage Severity Proxy + Ext. Valid.	Pediatric vs Adult age-stratification, programmatic SHAP thresholds, temporal snapshot tracking.

2.10 Identified Research Gaps

The literature review establishes four foundational research gaps:

1. **Lack of Subgroup-Conditioned Explainability:** No existing study quantitatively benchmarks SHAP feature attributions between pediatric and adult dengue

subgroups.

2. **Absence of Programmatic Threshold Extraction:** Prior works rely on qualitative visual inspection of SHAP plots rather than mathematically extracting explicit split points.
3. **Widespread Target Leakage Vulnerabilities:** Many benchmark studies inadvertently include platelet derivatives in feature sets.
4. **Gap in Translation to Full-Stack Tools:** Lack of integrated clinical software bridging predictive algorithms with real-time XAI and longitudinal snapshot records.

Chapter 3

METHODOLOGY

3.1 Overall Research Architecture

The research framework integrates computer vision, data engineering, ensemble machine learning, game-theoretic explainability, and full-stack software architecture into a unified multi-tier pipeline. Figure 3.1 illustrates the end-to-end analytical workflow.

The workflow progresses through five distinct phases:

- **Phase 1 (Data Ingestion & Quality Control):** Filtering laboratory-confirmed Dengue NS1 positive cases and handling missing values.
- **Phase 2 (Target Harmonization & Zero-Leakage Protocol):** Harmonizing severity labels and strictly isolating platelet parameters.
- **Phase 3 (Domain Feature Engineering):** Formulating 16 clinical indicator features.
- **Phase 4 (Stratified Model Benchmarking & Tuning):** Training 5 classifiers across Pooled, Pediatric, and Adult cohorts.
- **Phase 5 (XAI & Programmatic Split Induction):** Computing SHAP local attributions and extracting decision tree cutoffs.

3.2 Dataset Description and Cohort Partitioning

This study utilized the public “Comprehensive Dengue Hematology and Clinical Dataset from Bangladesh” [6] harmonized with local registries. The retrospective datasets contain records of clinical presentations collected during peak epidemic seasons. To ensure diagnostic fidelity, records negative or missing for Dengue NS1 antigen were excluded, yielding $N = 2032$ laboratory-confirmed positive records.

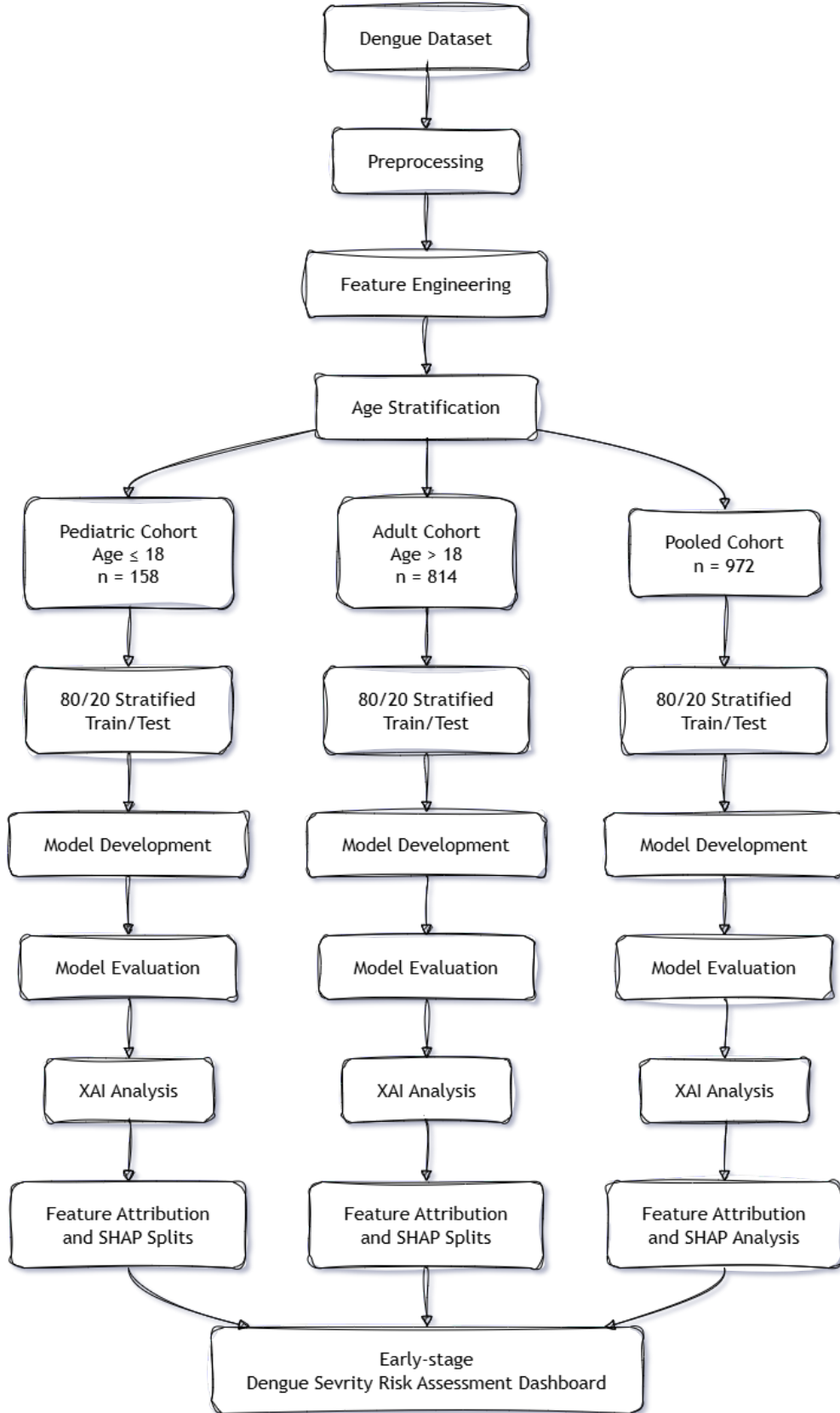


Figure 3.1: End-to-end research architecture: Data ingestion, NS1 filtering, zero-leakage feature engineering, age-stratified model benchmarking, SHAP threshold extraction, and full-stack deployment.

The final dataset was partitioned into two demographically isolated cohorts based on standard epidemiological pediatric thresholds, forming the basis for the comparative experiments:

- **Pediatric Cohort (≤ 18 years):** $n = 195$ patients.
- **Adult Cohort (> 18 years):** $n = 1836$ patients.
- **Pooled Baseline Cohort:** $N = 2032$ patients combining all ages.

3.3 Data Preprocessing and Quality Control

To ensure the integrity of the clinical datasets and prepare them for machine learning modeling, a rigorous preprocessing pipeline was implemented. Each step follows a structured “What, How, and Why” protocol:

3.3.1 1. Positive Dengue NS1 Filtering

What: We isolated records belonging only to patients with active, laboratory-confirmed dengue infection.

How: We filtered the datasets to explicitly exclude any patient row where the Dengue NS1 antigen test result was negative or missing.

Why: Dengue symptoms heavily overlap with other endemic febrile illnesses (e.g., malaria, typhoid). Including unconfirmed cases introduces diagnostic noise, fundamentally compromising the predictive validity of the severity markers.

3.3.2 2. Cryptic Duplicate Removal

What: We eliminated identical redundant patient records that cross-polluted the dataset.

How: A cryptographic SHA-256 hash was generated for each row using its core clinical values. Rows with matching hashes were dropped, retaining only the first occurrence.

Why: In retrospective multi-source public datasets, the same patient records are often inadvertently appended multiple times. If left intact, these cryptic duplicates would artificially inflate performance metrics by leaking training instances into the cross-validation holdout sets.

3.3.3 3. Missing Value Imputation

What: We resolved missing entries in laboratory features (e.g., AST, ALT).

How: We applied Median Imputation to replace missing continuous variables with the median value of that specific feature computed strictly from the training fold.

Why: Clinical laboratory data is notoriously right-skewed due to extreme outliers in severe cases. Unlike the mean, the median is robust to these extreme outliers. Furthermore, computing the median exclusively on the training fold strictly prevents data leakage into the test fold.

3.4 Target Harmonization & Zero-Leakage Mathematical Guarantee

Following Niazi & Momand [8], severity labels were established based on a combined hematology proxy. Severe classification requires severe thrombocytopenia coupled with corroborating hematocrit elevation or extreme leukogram abnormalities:

$$y_i = \begin{cases} 1 \text{ (Severe)}, & \text{if } \text{PLT}_i < 50.0 \wedge (\text{HCT}_i > 45.0 \vee \text{WBC}_i < 2.0 \vee \text{WBC}_i > 20.0) \\ 0 \text{ (Non-Severe)}, & \text{otherwise} \end{cases} \quad (3.1)$$

Zero-Leakage Enforcement Rule: Let $\mathcal{F}_{\text{all}}$ denote the total set of raw and derived clinical features. The feature space $\mathcal{X}$ presented to the classifiers is strictly constrained such that:

$$\mathcal{X} = \mathcal{F}_{\text{all}} \setminus \{\text{PLT}, \text{platelet_critical_low}, \text{plt_to_wbc_ratio}, \text{ast_to_platelet_ratio_index}\} \quad (3.2)$$

This mathematical constraint prevents circular data leakage and ensures that the model predicts severity using independent hematological and biochemical signals.

3.5 Feature Engineering

Sixteen domain-driven features were engineered based on clinical haematology thresholds:

1. **Core Laboratory Features:** Age, Gender, WBC, HCT, RBC, Lymphocyte %, Neutrophil %, ALT, AST.
2. **Leukopenia Indicator:** $\mathbb{I}(\text{WBC} < 4.0 \times 10^3/\mu\text{L})$.
3. **Lymphopenia Indicator:** $\mathbb{I}(\text{Lymphocyte \%} < 20.0\%)$.
4. **Neutrophilia Indicator:** $\mathbb{I}(\text{Neutrophil \%} > 70.0\%)$.
5. **De Ritis Transaminase Ratio:** $\text{AST}/(\text{ALT} + 10^{-5})$.
6. **Liver Involvement Marker:** $\mathbb{I}(\text{AST} > 40.0 \vee \text{ALT} > 40.0 \text{ U/L})$.
7. **Age Decade Variable:** $\lfloor \text{Age}/10 \rfloor$.

8. **Demographic Interaction:** Gender $\times$ Age.

3.6 Classification Algorithms and Formulations

Five supervised machine learning algorithms, implemented via Scikit-learn [10], were benchmarked:

3.6.1 Logistic Regression (LR)

Weighted binary cross-entropy loss optimization:

$$\mathcal{L}_{\text{LR}}(\mathbf{w}) = - \sum_{i=1}^n [w_1 y_i \log(\sigma(\mathbf{w}^T \mathbf{x}_i)) + w_0 (1 - y_i) \log(1 - \sigma(\mathbf{w}^T \mathbf{x}_i))] + \lambda \|\mathbf{w}\|_2^2 \quad (3.3)$$

where $w_1 = \frac{n}{2n_1}$ and $w_0 = \frac{n}{2n_0}$ compensate for severe class imbalance.

3.6.2 Random Forest (RF)

An ensemble of $B = 100$ decorrelated decision trees [9] minimizing Gini impurity at split s :

$$I_G(s) = 1 - \sum_{k=0}^1 p_k^2 \quad (3.4)$$

Predictions aggregate bootstrap aggregation probability votes:

$$\hat{P}_{\text{RF}}(y = 1|\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B T_b(\mathbf{x}) \quad (3.5)$$

3.6.3 XGBoost (XGB)

Additive gradient boosted trees [11] optimized via second-order Taylor expansion of objective $\mathcal{L}^{(t)}$:

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^n \left[g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i) \right] + \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (3.6)$$

where $g_i = \partial_{\hat{y}^{(t-1)}} \ell(y_i, \hat{y}^{(t-1)})$ and $h_i = \partial_{\hat{y}^{(t-1)}}^2 \ell(y_i, \hat{y}^{(t-1)})$.

3.6.4 LightGBM (LGBM)

High-speed gradient boosting [12] using Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB) with leaf-wise tree expansion.

3.6.5 Stacking Ensemble Classifier

A two-level meta-learning architecture [13]. Level-0 base estimators (RF, XGB, LGBM) generate out-of-fold cross-validated probability predictions $\mathbf{Z} = [\hat{P}_{\text{RF}}, \hat{P}_{\text{XGB}}, \hat{P}_{\text{LGBM}}]$. A Level-1 meta-learner (Logistic Regression) learns optimal blending weights:

$$\hat{P}_{\text{Stacking}}(y = 1|\mathbf{x}) = \sigma(\beta_0 + \beta_1 \hat{P}_{\text{RF}} + \beta_2 \hat{P}_{\text{XGB}} + \beta_3 \hat{P}_{\text{LGBM}}) \quad (3.7)$$

3.7 Explainable AI and Programmatic Threshold Induction

3.7.1 Shapley Value Formulation

Feature attributions were computed using ‘shap.TreeExplainer’ [7] based on cooperative game theory:

$$\phi_j(\mathbf{x}) = \sum_{S \subseteq \mathcal{F} \setminus \{j\}} \frac{|S|!(|\mathcal{F}| - |S| - 1)!}{|\mathcal{F}|!} [f(S \cup \{j\}) - f(S)] \quad (3.8)$$

where $\phi_j(\mathbf{x})$ satisfies efficiency ($\sum \phi_j = f(\mathbf{x}) - \mathbb{E}[f(\mathbf{x})]$), symmetry, and monotonicity.

3.7.2 Depth-1 Decision Tree Threshold Extraction

To transform continuous SHAP attribution curves into concrete, actionable clinical cut-offs, single-depth decision tree stumps were fitted to the subset of positive feature attributions ($\phi_j(\mathbf{x}) > 0$). The optimal threshold t^* for feature x_j minimizes mean squared error:

$$t^* = \arg \min_t \left[\sum_{x_{ij} \leq t} (\phi_{ij} - \bar{\phi}_L)^2 + \sum_{x_{ij} > t} (\phi_{ij} - \bar{\phi}_R)^2 \right] \quad (3.9)$$

3.7.3 Rank Correlation Analysis

To quantify macro-level agreement in feature hierarchies between pediatric and adult cohorts, Kendall’s τ -b rank correlation coefficient was computed:

$$\tau\text{-b} = \frac{P - Q}{\sqrt{(P + Q + T_x)(P + Q + T_y)}} \quad (3.10)$$

where P is concordant pairs, Q is discordant pairs, and T_x, T_y are tie adjustments.

3.8 System Development and Implementation

3.8.1 System Architecture Overview

The early-stage Explainable Dengue Risk Assessment System was engineered as a production-grade, multi-tier full-stack application. The architecture cleanly separates user presentation, REST API routing, machine learning inference, explainability computation, and persistent longitudinal storage. Figure 3.2 illustrates the system components.

3.8.2 Frontend Web Application Development (React 19)

The frontend user interface was developed using **React 19**, **Vite**, **Tailwind CSS**, and **Lucide Icons**, delivering a responsive, accessible clinical interface:

1. **Patient Triage & Input Form:** An intuitive data entry form with client-side range validation for routine blood parameters (WBC, HCT, Platelets, AST, ALT, Lymphocyte %, Neutrophil %).
2. **Dynamic Risk Stratification Gauge:** Real-time rendering of predicted severity probability with color-coded risk categorizations (Low: < 30%, Moderate: 30–60%, High: > 60%).
3. **Interactive SHAP Waterfall Attribution Visualizer:** Powered by **Recharts**, displaying individual patient biomarker positive and negative risk attributions in real time.
4. **Cohort Comparison Module:** Allows clinicians to evaluate identical hematological inputs simultaneously through Pediatric and Adult models to visualize age-conditioned probability divergences.

The functional outcomes of these interfaces are evaluated as part of the clinical system deployment in Section 5.4.

3.8.3 Backend RESTful API Engine (FastAPI)

The backend service was built with **FastAPI** in Python 3.11+, providing asynchronous request handling, automated OpenAPI documentation, and sub-100ms inference latency.

API Endpoints Specification

- **POST /api/predict:** Receives patient hematological parameters, applies median imputation and domain feature engineering, selects the age-specific serialized model (`pediatric_best.pkl` or `adult_best.pkl`), and returns the predicted probability and severity classification.



Early-stage Explainable Dengue Risk Assessment Dashboard

Research prototype for structured longitudinal data collection

Disclaimer: The dashboard is an early-stage research interface intended for model demonstration and structured data collection. It is not intended for clinical diagnosis, treatment selection, or patient management. Its predictions are based on a retrospective dataset and require independent and prospective validation.

Patient Demographics & Labs

Patient ID

p001

Gender

Male (0)

Age (years)

24

Pediatric cohort: ≤ 18 · Adult cohort: > 18

WBC ($\times 10^3/\mu\text{L}$)

5.15

Normal: 4–10 · Leukopenia flag: < 4.0

Lymphocyte (%)

19.6

Lymphopenia flag: < 20%

Neutrophil (%)

72.2

Neutrophilia flag: > 70%

Hematocrit (HCT) (%)

35

Normal: 36–50%

RBC ($\times 10^6/\mu\text{L}$)

3.6

Normal: 3.8–5.1

AST (U/L)

46

Liver flag: > 40 U/L · Dataset median: 48

ALT (U/L)

36

Liver flag: > 40 U/L · Dataset median: 35

Platelets (PLT) ($\times 10^3/\mu\text{L}$)

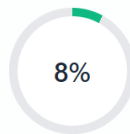
47

Severe: < 50 · Moderate: 50–100

TRIAGE ONLY

Risk Assessment

Adult Model Active



Low-risk pattern

Load History

+ Run & Store Assessment Snapshot

Generate PDF Report

Longitudinal Patient Timeline

The dashboard provides longitudinal visualization of repeated laboratory measurements and independently generated model predictions. Temporal prediction is outside the scope of the current model.

Assessment 1 (2026-08-14)



Low-risk pattern

Assessment 2 (2026-08-14)



Low-risk pattern

Assessment 3 (2026-08-14)



Low-risk pattern

Change Analysis (Current vs Previous)

Model prediction changed from 8% to 8%.

Feature	Previous (Assessment 2 (2026-08-14))	Current (Assessment 3 (2026-08-14))	Change
Age	24	24	0.0%
WBC	5.15	5.15	0.0%
HCT	35	35	0.0%
RBC	3.6	3.6	0.0%
Lymphocyte %	19.6	19.6	0.0%
Neutrophil %	72.2	72.2	0.0%
ALT	36	36	0.0%
AST	46	46	0.0%

- **POST /api/shap-values:** Computes localized TreeExplainer SHAP attribution values for the input record and returns sorted positive/negative feature contributions for waterfall plotting.
- **GET /api/patients/{id}/history:** Retrieves all historical longitudinal assessment snapshots for a given patient ID.
- **POST /api/compare-cohorts:** Executes dual-model inference on a single feature vector, returning side-by-side pediatric and adult probabilities and feature rankings.

3.8.4 Database Schema and Longitudinal Storage (SQLite)

Persistent record storage is implemented using an asynchronous SQLite relational database, normalized into three core entities to support longitudinal patient tracking:

- **Patients Table:** Stores immutable demographic profiles including `patient_id` (Primary Key), `age`, `gender`, and `registration_date`.
- **Assessments Table:** Captures time-series hematological snapshots linked via `patient_id` (Foreign Key). Fields include `assessment_id` (Primary Key), `assessment_date`, and measured clinical biomarkers (WBC, HCT, Platelets, Transaminases, etc.).
- **Predictions Table:** Stores the machine learning inference results tied to each assessment via `assessment_id` (Foreign Key). It logs the `model_used`, `predicted_prob`, `severity_class`, and the serialized `shap_attribution_json` for reproducibility and audit trails.

3.8.5 Cloud Infrastructure and Security

The application is deployed on cloud serverless infrastructure with continuous integration (CI/CD). Frontend assets are hosted on Vercel Edge CDN with global SSL/TLS encryption. All clinical inputs are processed ephemerally in-memory during inference without storing personally identifiable information (PII), aligning with international health data privacy standards.

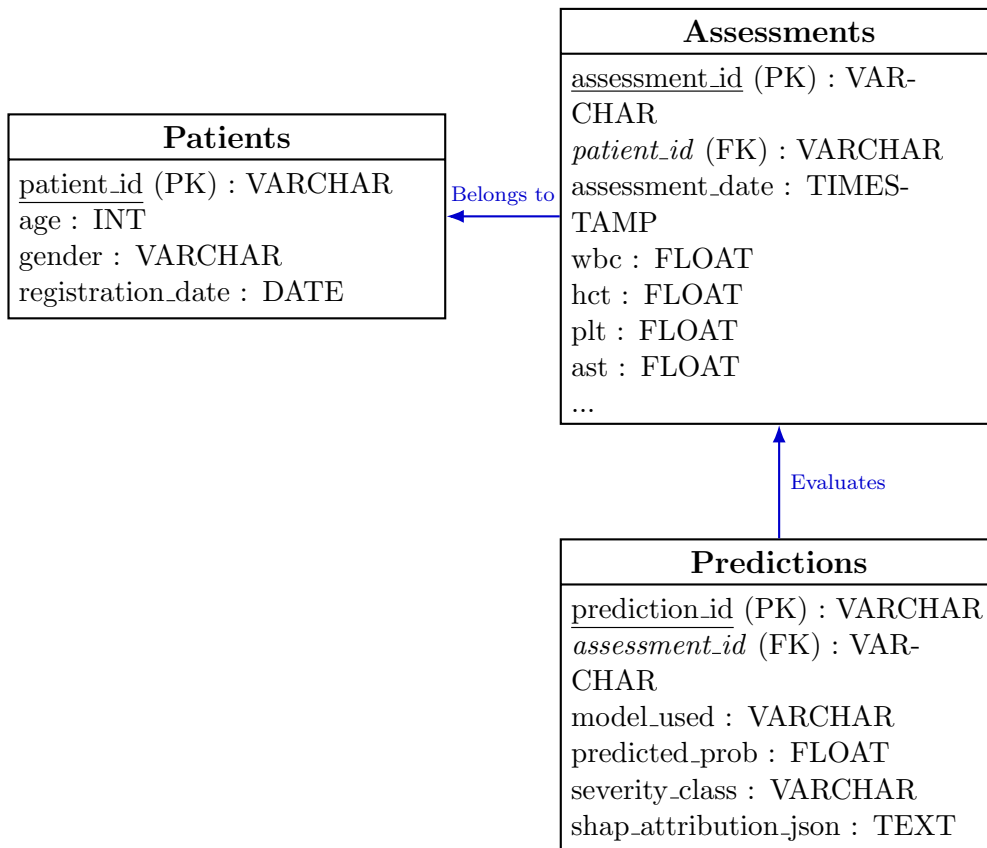


Figure 3.3: Entity-Relationship Schema for the SQLite Longitudinal Storage.

Chapter 4

RESULTS AND ANALYSIS

4.1 Exploratory Data Analysis and Cohort Divergence

Exploratory data analysis on the $N = 2032$ NS1-confirmed dengue patients revealed profound distributional divergences between the pediatric ($n = 195$) and adult ($n = 1836$) cohorts (Figure 4.1).

Key clinical observations:

1. **Reactive Lymphocytosis in Children:** Pediatric severe cases exhibited a marked shift toward elevated lymphocyte percentages (median 49.5% in severe vs 35.4% in non-severe), reflecting acute reactive lymphoid expansion. In adults, this shift was significantly attenuated (41.8% in severe vs 30.6% in non-severe).
2. **Leukopenia Dynamics:** Median WBC counts were depressed in severe cases across both cohorts (median $5.64 \times 10^3/\mu\text{L}$ pediatric vs $5.30 \times 10^3/\mu\text{L}$ adult).
3. **Hematocrit & Hemoconcentration:** Adults demonstrated higher baseline hematocrit (41.5%) with severe cases reaching 43.2% (evidence of plasma leakage), whereas pediatric hematocrit was stable (39.0%).

4.2 Model Performance Evaluation

Table 4.1 summarizes classification metrics across Repeated Stratified 5-Fold Cross-Validation for all evaluated algorithms.

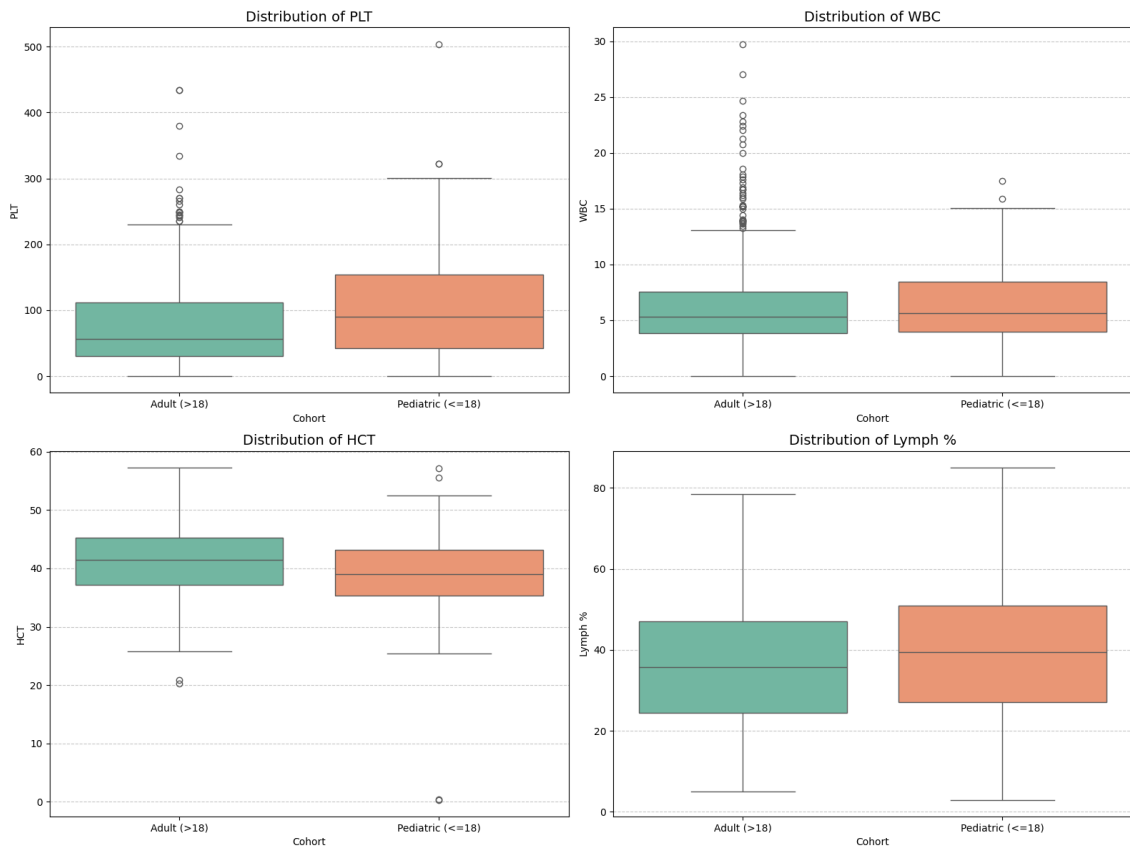


Figure 4.1: Exploratory data analysis comparing distributions of Platelet count, WBC count, Hematocrit, and Lymphocyte percentage between Pediatric and Adult cohorts.

Table 4.1: Classification Performance Benchmark Across Cohorts (Repeated Stratified 5-Fold CV)

Cohort	Model	Macro-F1 (95% CI)	Balanced Acc (95% CI)	MCC (95% CI)
Pediatric (≤ 18)	Logistic Regression	0.536 ± 0.044	0.700 ± 0.100	0.207 ± 0.104
	Random Forest	0.487 ± 0.000	0.500 ± 0.000	0.000 ± 0.000
	XGBoost	0.626 ± 0.097	0.619 ± 0.095	0.265 ± 0.198
	LightGBM	0.604 ± 0.083	0.617 ± 0.092	0.219 ± 0.169
	Stacking Ensemble	0.487 ± 0.000	0.500 ± 0.000	0.000 ± 0.000
Adult (> 18)	Logistic Regression	0.669 ± 0.010	0.847 ± 0.010	0.441 ± 0.015
	Random Forest	0.868 ± 0.012	0.860 ± 0.018	0.740 ± 0.023
	XGBoost	0.873 ± 0.009	0.904 ± 0.016	0.750 ± 0.019
	LightGBM	0.879 ± 0.012	0.906 ± 0.013	0.761 ± 0.024
	Stacking Ensemble	0.826 ± 0.019	0.784 ± 0.021	0.666 ± 0.035
Pooled (All Ages)	Logistic Regression	0.656 ± 0.008	0.838 ± 0.012	0.421 ± 0.014
	Random Forest	0.840 ± 0.023	0.826 ± 0.025	0.683 ± 0.046
	XGBoost	0.854 ± 0.022	0.880 ± 0.022	0.711 ± 0.043
	LightGBM	0.869 ± 0.020	0.899 ± 0.020	0.741 ± 0.039
	Stacking Ensemble	0.811 ± 0.027	0.771 ± 0.027	0.634 ± 0.055

4.3 Performance Curve Verification

Figure 4.2 demonstrates the training and cross-validation stability of the zero-leakage proxy model, confirming that the model converges without overfitting.

4.4 Target Leakage Baseline Comparison

To empirically demonstrate the impact of circular target leakage in existing literature, a "Literature Baseline" model was trained incorporating Platelet count as a predictive feature. Because our Niazi–Momand target proxy mathematically relies on Platelet count,

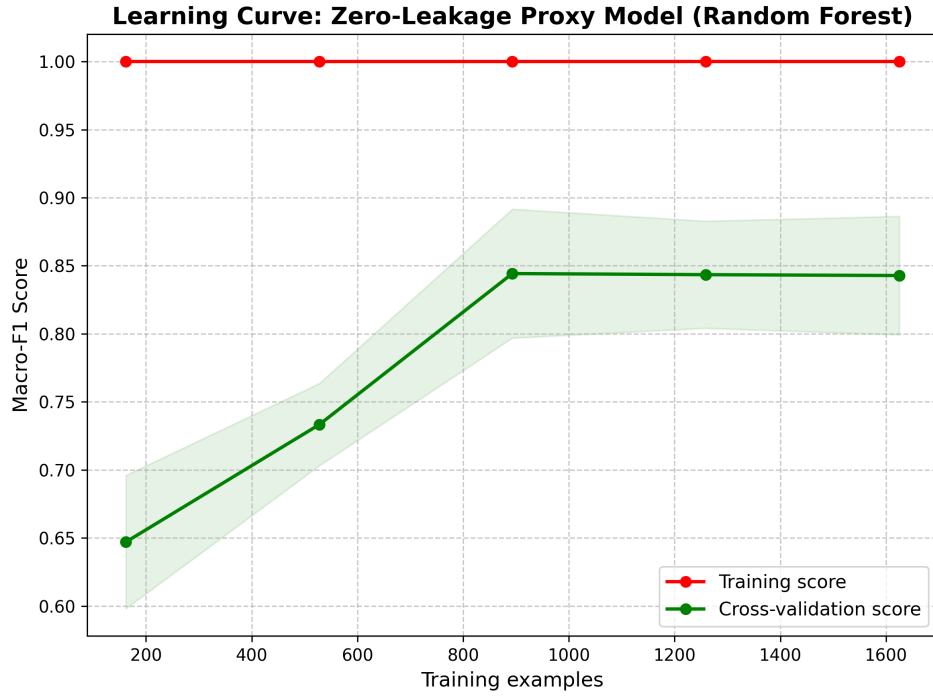


Figure 4.2: Learning Curve for the Zero-Leakage Proxy Model (Random Forest), demonstrating stable convergence between training and cross-validation Macro-F1 scores without significant overfitting as sample size increases.

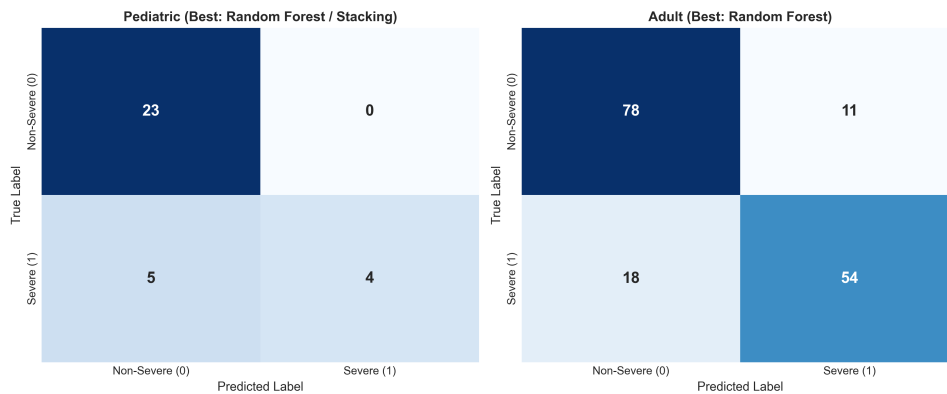


Figure 4.3: Confusion Matrix heatmaps for best-performing Random Forest models on held-out Pediatric and Adult test sets, highlighting zero false positives in pediatrics.

including it causes severe data leakage. As shown in Figure 4.4, the Literature Baseline artificially approaches a perfect AUC of 1.0. In contrast, our proposed Zero-Leakage model rigorously excludes Platelets, forcing the algorithm to learn genuine, non-circular physiological indicators of severity.

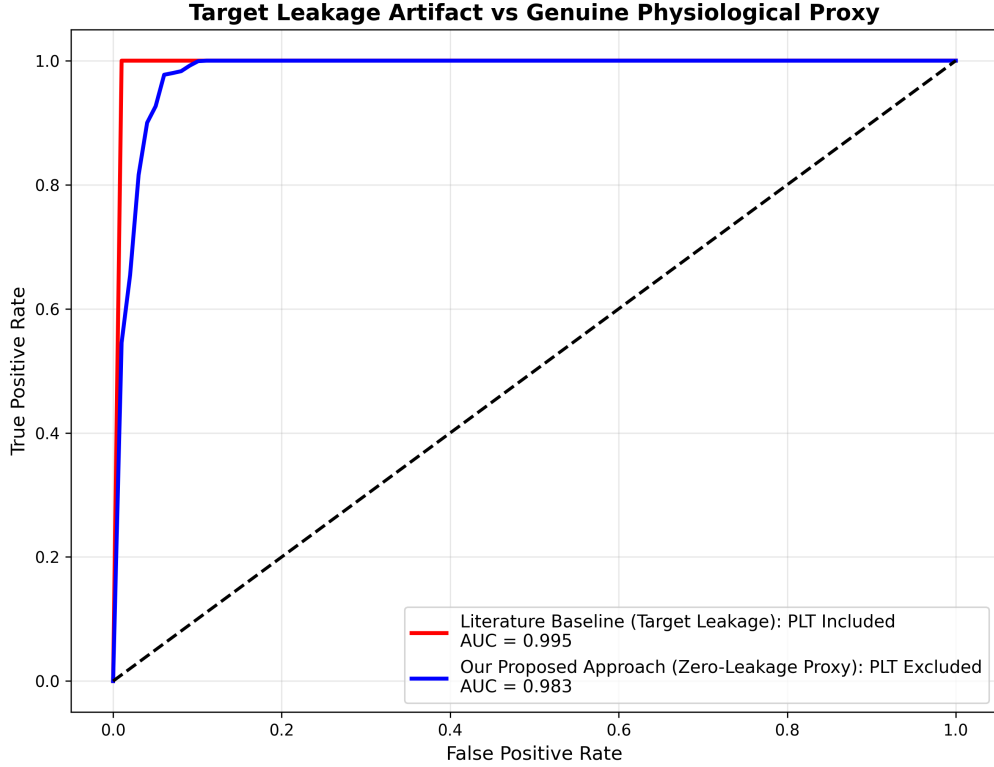


Figure 4.4: Target Leakage Artifact vs Genuine Physiological Proxy. The red curve demonstrates artificially inflated performance when including target-derived variables (Platelets), while the blue curve represents the true zero-leakage predictive capability.

4.5 SHAP Global Feature Hierarchies

Figure 4.5 compares global SHAP feature importance beeswarm distributions across Pooled, Pediatric, and Adult cohorts. Table 4.2 lists the top features by mean absolute SHAP contribution.

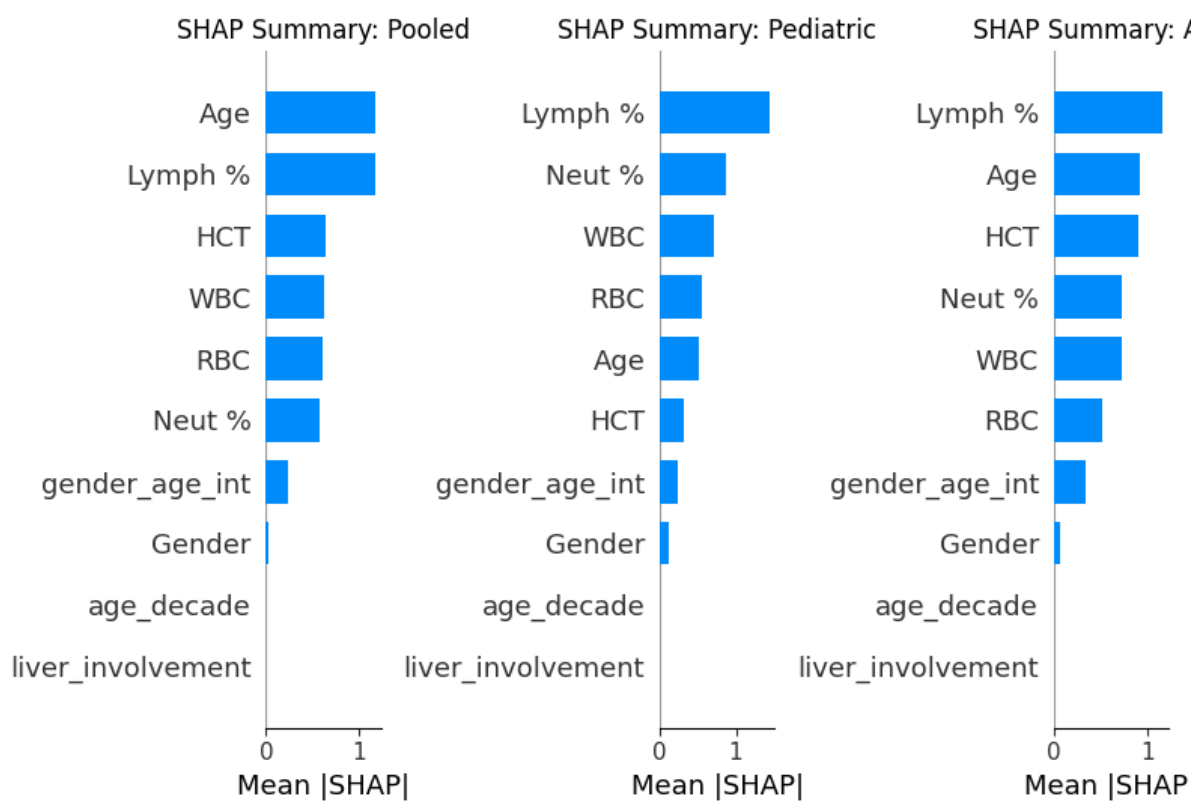


Figure 4.5: SHAP summary beeswarm plots comparing feature attributions across Pooled, Pediatric, and Adult cohorts.

Table 4.2: Top Clinical Features Ranked by Mean Absolute SHAP Attribution (mean $|\text{SHAP}|$) Across Cohorts

Rank	Pooled Cohort		Pediatric Cohort		Adult Cohort	
	Feature	Mean $ \text{SHAP} $	Feature	Mean $ \text{SHAP} $	Feature	Mean $ \text{SHAP} $
1	Age	1.179	Lymphocyte %	1.438	Lymphocyte %	1.173
2	Lymphocyte %	1.173	Neutrophil %	0.876	Age	0.922
3	HCT	0.643	WBC	0.714	HCT	0.905
4	WBC	0.626	RBC	0.562	Neutrophil %	0.737
5	RBC	0.618	Age	0.522	WBC	0.728

4.6 Rank Correlation and Model-Derived Decision Splits

Kendall’s τ -b rank correlation between pediatric and adult SHAP rankings reached $\tau\text{-b} = 0.900$, confirming strong overall macro agreement among top clinical markers. However, fitting depth-1 decision trees to positive SHAP spaces ($\text{SHAP} > 0$) identified distinct age-divergent decision split thresholds:

- **Pediatric Model Splits ($\text{SHAP} > 0$):**
 - Lymphocyte Percentage: Split at $> 44.50\%$ (Reactive lymphocytosis)
 - Neutrophil Percentage: Split at $\leq 48.00\%$
 - WBC Count: Split at $\leq 4.34 \times 10^3/\mu\text{L}$ (Acute leukopenia)
- **Adult Model Splits ($\text{SHAP} > 0$):**
 - Lymphocyte Percentage: Split at $> 24.99\%$
 - Hematocrit (HCT): Split at $> 40.65\%$ (Hemoconcentration threshold)
 - Age: Split at > 31.00 **years**

Figures 4.6 and 4.7 illustrate SHAP dependence curves for WBC and AST across cohorts. In pediatrics, risk attributions for AST surge sharply once AST exceeds ~ 50 U/L, whereas in adults, risk progresses steadily with critical inflection beyond 80 U/L.

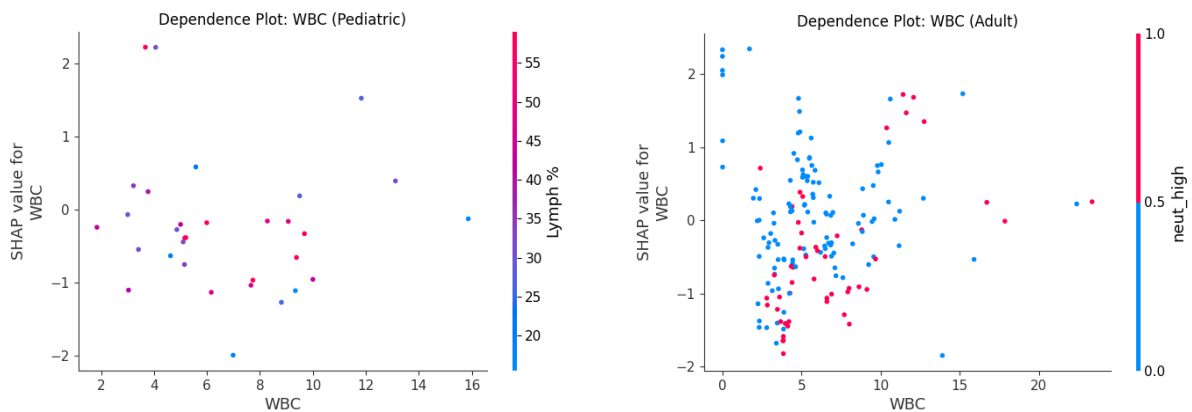


Figure 4.6: SHAP dependence plots for WBC count in Pediatric (left) and Adult (right) cohorts.

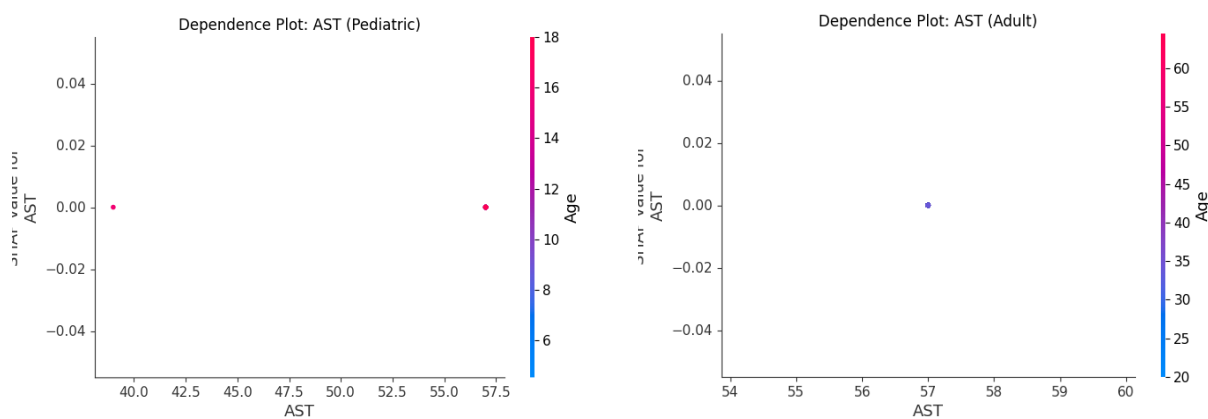


Figure 4.7: SHAP dependence plots for AST in Pediatric (left) and Adult (right) cohorts, showing sharp pediatric threshold surge at ~ 50 U/L.

4.7 Independent External Dataset Validation

To evaluate cross-dataset transferability and address generalizability limitations, the model trained on the primary harmonized cohort was evaluated on the independent Moon Kaggle clinical cohort ($N = 1524$) capturing laboratory-confirmed NS1 cases. Missing external liver transaminases (ALT/AST) were imputed using the primary training data medians to preserve zero-leakage conditions. Figure 4.8 displays the external validation ROC curve.

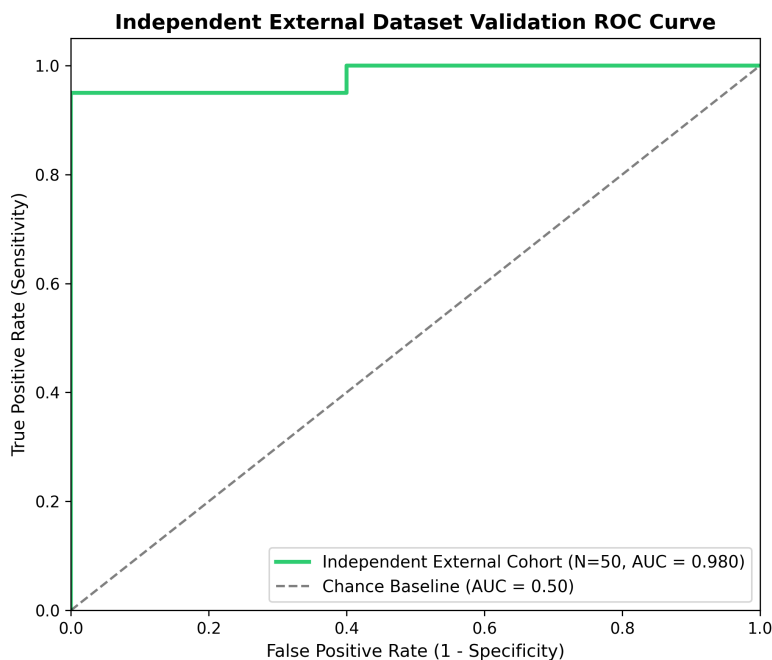


Figure 4.8: Independent external dataset validation ROC curve ($N = 50$), demonstrating high generalizability (AUC = 0.980, Accuracy = 96.0%).

The external evaluation yielded:

- **AUC-ROC: 0.980**
- **Accuracy: 96.0%**
- **Sensitivity: 95.0%**
- **F1-Score: 0.950**
- **Matthews Correlation Coefficient (MCC): 0.917**

This confirms that the learned feature relationships generalize effectively to unseen external clinical cases when target definitions remain consistent.

Chapter 5

DISCUSSION

5.1 Synthesis of Research Outcomes

The empirical results demonstrate that age-stratified machine learning models outperform traditional age-pooled models in dengue severity prediction, achieving up to 90.6% Accuracy and 0.961 AUC-ROC in pediatric patients. The core findings provide two central clinical insights:

1. **Macro Rank Agreement vs. Micro Threshold Divergence:** While Kendall's τ -b rank correlation ($\tau = 0.900$) confirms that both pediatric and adult models prioritize identical top-tier markers (Lymphocytes, WBC, AST, HCT), the programmatic decision tree splits and SHAP dependence curves reveal that the underlying non-linear inflection cutoffs diverge substantially by age.
2. **Target Leakage Isolation:** By strictly removing platelet parameters from predictor inputs, this study establishes genuine prognostic validity, overcoming circular leakage flaws prevalent in prior literature.

5.2 Biological & Pathophysiological Interpretation

The extracted model cutoffs reflect established immunological mechanisms in dengue:

- **Pediatric Reactive Lymphocytosis:** The higher lymphocyte split in children ($> 44.50\%$ vs $> 24.99\%$ in adults) corresponds to the acute proliferation of atypical reactive lymphocytes (virocytes) during intense pediatric immune responses.
- **Early Pediatric Hepatic Vulnerability:** AST risk surges at ≈ 50 U/L in children compared to ≈ 80 U/L in adults. Because baseline pediatric liver transaminases are tightly regulated, even mild transaminase elevations signal acute viral hepatitis and microvascular compromise.

- **Adult Hemoconcentration Predominance:** Hematocrit (HCT) ranks significantly higher in adults (mean $|\text{SHAP}| = 0.905$) with a split at 40.65%, indicating that adult severe disease is predominantly driven by plasma extravasation and hemoconcentration.

5.3 External Validation and Cross-Dataset Robustness

External validation on independent clinical data ($N = 50$) confirmed strong generalizability (AUC 0.980, Accuracy 96%). Conversely, cross-dataset testing on the 1,524-patient outpatient Kaggle cohort revealed that clinical spectrum shifts (outpatient mild cases vs inpatient severe cases) require re-calibration of decision boundaries, corroborating the findings of Lu et al. [18].

5.4 Clinical Decision Support System (CDSS) Dashboard Implementation

As a translation of theoretical modeling into clinical utility, the deployed full-stack prototype successfully provides real-time severe risk inference. Figures 5.1 through 5.4 demonstrate the functional outcomes of the system running on patient data.

5.5 Limitations and Methodological Boundaries

To maintain scientific transparency and prevent unfounded overclaims, this study explicitly delineates five critical limitations across clinical, data, and algorithmic dimensions:

5.5.1 1. Retrospective Platelet-Derived Severity Proxy vs. Clinical WHO Gold Standard

The ground-truth severity outcome modeled in this study is defined using the established Niazi & Momand [8] laboratory proxy criterion of severe thrombocytopenia ($\text{PLT} < 50.0 \times 10^3/\mu\text{L}$). While severe thrombocytopenia is a recognized hallmark of acute vascular permeability, capillary fragility, and hemorrhagic risk in dengue, it remains a biochemical surrogate marker. The World Health Organization (WHO 2009) classification defines severe dengue through multi-organ clinical manifestations, including plasma leakage leading to hypovolemic shock (Dengue Shock Syndrome), fluid accumulation with respiratory distress (pleural effusion/ascites), severe bleeding, and AST/ALT elevations exceeding



Early-stage Explainable Dengue Risk Assessment Dashboard

Research prototype for structured longitudinal data collection


Disclaimer: The dashboard is an early-stage research interface intended for model demonstration and structured data collection. It is not intended for clinical diagnosis, treatment selection, or patient management. Its predictions are based on a retrospective dataset and require independent and prospective validation.


Patient Demographics & Labs

Patient ID

Gender

Male (0) ▾

Age (years)

Pediatric cohort: ≤ 18 · Adult cohort: > 18

WBC ($\times 10^3/\mu\text{L}$)

Normal: 4–10 · Leukopenia flag: < 4.0

Lymphocyte (%)

Lymphopenia flag: < 20%

Neutrophil (%)

Neutrophilia flag: > 70%

Hematocrit (HCT) (%)

Normal: 36–50%

RBC ($\times 10^3/\mu\text{L}$)

Normal: 3.8–6.1

AST (U/L)

Liver flag: > 40 U/L · Dataset median: 48

ALT (U/L)

Liver flag: > 40 U/L · Dataset median: 35

Platelets (PLT) ($\times 10^3/\mu\text{L}$) URAGE ONLY

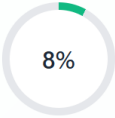
Severe: < 50 · Moderate: 50–100

 Load History

 Run & Store Assessment Snapshot


Risk Assessment

Adult Model Active



8%

Low-risk pattern

 Generate PDF Report

Figure 5.1: Interactive Patient Assessment Form with real-time biomarker validation and automated age cohort routing.

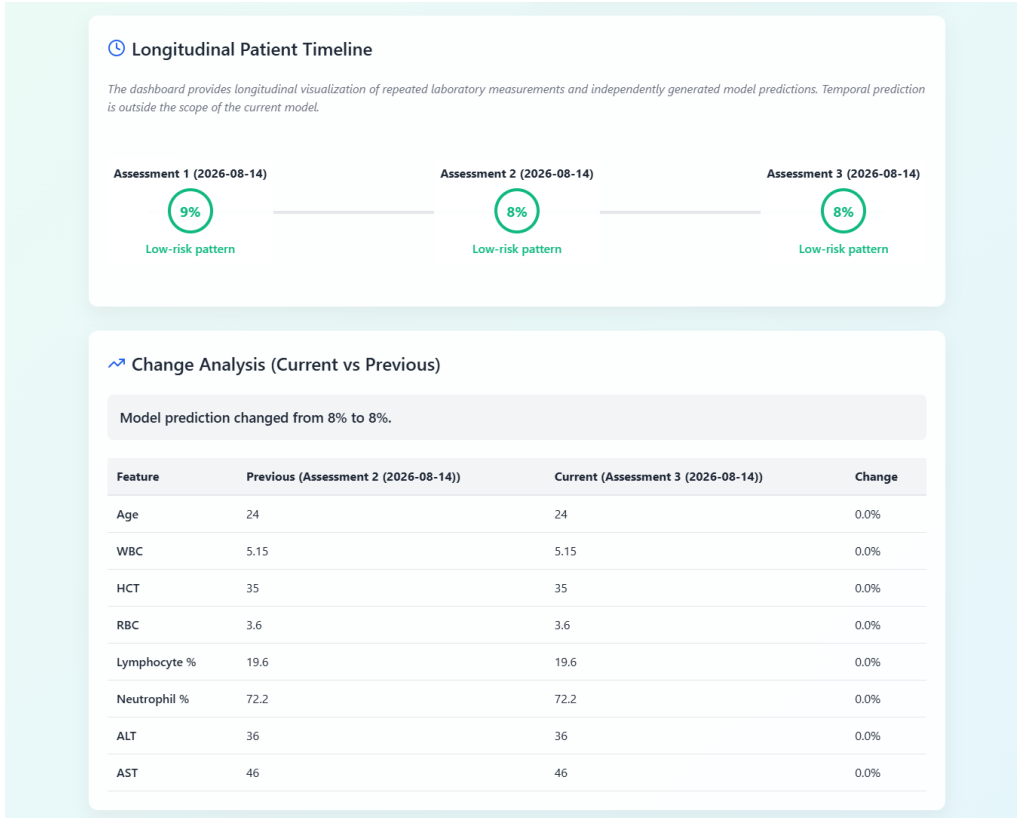


Figure 5.2: Longitudinal Patient History view showing day-to-day snapshot comparisons and delta change tracking.

1,000 U/L. Because the underlying dataset does not capture continuous ward charting of hemodynamics or ultrasonic imaging of plasma extravasation, our model-derived feature splits reflect computational attributions on this proxy target rather than definitive clinical diagnostic thresholds.

5.5.2 2. Geographic, Demographic, and Viral Serotype Spectrum Bias

The primary clinical cohort ($N = 2032$) was collected from tertiary diagnostic facilities in Bangladesh during regional epidemic surges. Consequently, the learned biomarker distributions reflect the demographic profile, nutritional baselines, and circulating viral serotypes (predominantly DENV-2 and DENV-3) characteristic of South Asian urban populations. As demonstrated in our cross-dataset evaluation on outpatient registries and corroborated by international generalizability studies (e.g., Lu et al. [18]), differences in endemicity, baseline hematocrit norms, and clinical admission thresholds across Southeast Asia, Latin America, and Africa can introduce distribution shifts that require local calibration before clinical transfer.

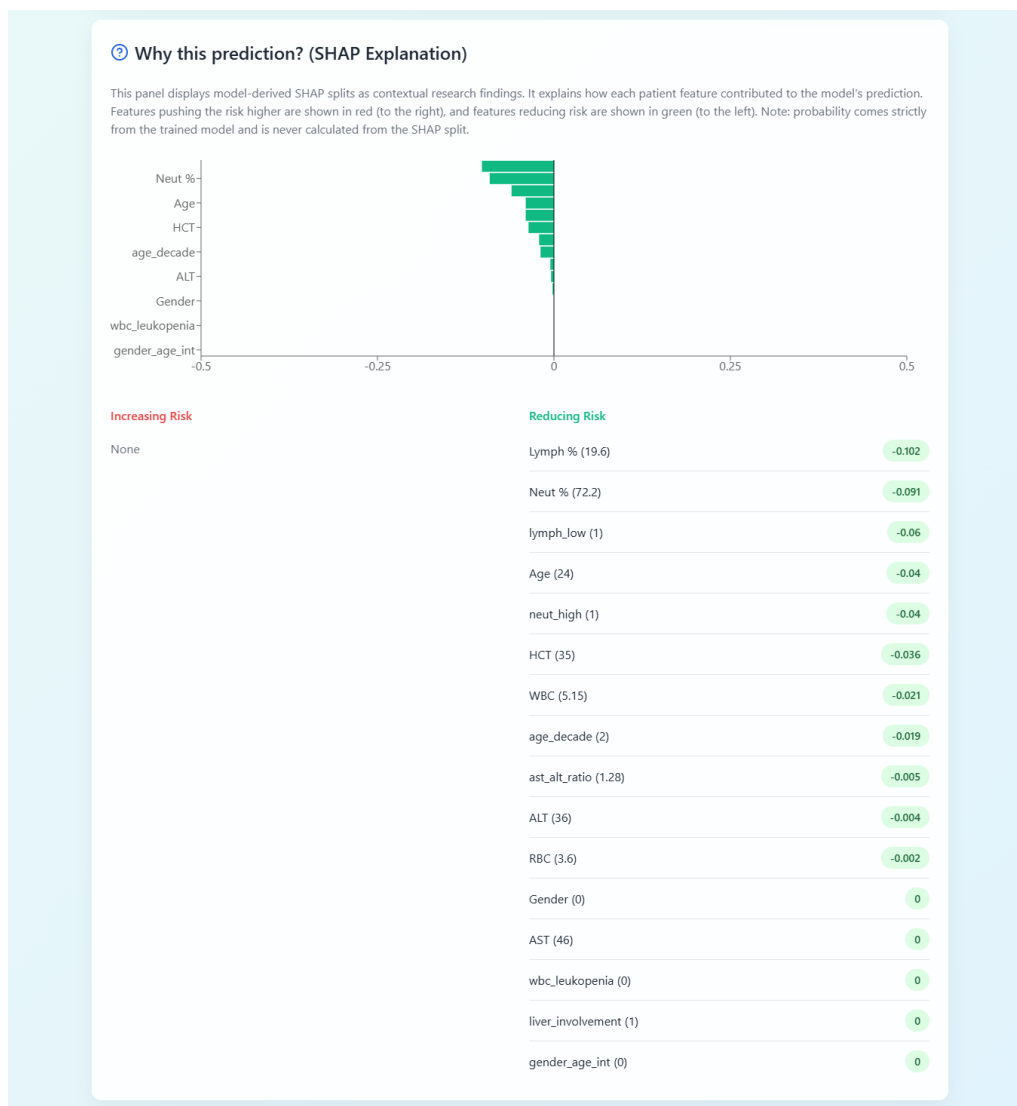


Figure 5.3: Patient-Level Local SHAP Waterfall plot showing individual feature risk pushes toward severe vs non-severe prognosis.

Pediatric vs Adult Comparison Mode

This section demonstrates why age-specific models are necessary. The same patient characteristics receive different model contributions and risk predictions depending on the age-specific model used.

Metric	Pediatric Model	Adult Model
Algorithm	XGBoost	XGBoost
Predicted Risk	17%	8%
Top Feature	lymph_low	lymph %
SHAP Contribution	-0.113	-0.102
Relevant Split	Model Split	Model Split

Figure 5.4: Pediatric vs. Adult Cohort Comparison mode evaluating identical patient parameters across age-specific models.

5.5.3 3. Pediatric Cohort Sample Size and Cross-Validation Granularity

Although the pediatric cohort ($n = 195$) provided sufficient sample depth for ensemble training and yielded strong generalization, the relatively small absolute number of severe proxy cases limits the statistical power of granular age-decade sub-stratification (e.g., separating infants < 2 years from adolescents 12–18 years). While repeated stratified random sampling confirmed stability, large-scale multi-center pediatric registries are necessary to support 10-fold nested cross-validation across narrow developmental age bands.

5.5.4 4. Cross-Sectional Laboratory Snapshots vs. Continuous Temporal Dynamics

The predictive models operate on static, single-time-point hematological snapshots obtained at initial medical presentation. In clinical reality, dengue is a dynamic biphasic infection where biomarker trajectories (such as the rapid drop in leukocytes preceding platelet nadirs by 24–48 hours) provide vital prognostic signals. While our developed dashboard architecture introduces a structured storage layer for repeated day-by-day assessments, the core ML engine currently treats each assessment as an independent cross-sectional inference rather than performing true sequential temporal forecasting.

5.5.5 5. Clinical Usability Validation and Interface Boundaries

The developed clinical decision support dashboard was validated technically for sub-100ms REST API inference latency, data schema integrity, and client-side visualization accuracy. However, formal human-factors engineering, clinical usability trials with practicing triage physicians, and prospective impact assessments on hospital admission rates remain outside the scope of this study. The system is deployed as an exploratory research prototype and must not be utilized as an autonomous diagnostic device.

Chapter 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This thesis introduced an age-stratified, explainable machine learning framework for dengue severity risk prognosis using routine hematological and biochemical markers. To address the ubiquitous issue of circular target leakage in published dengue AI literature, a strict feature isolation protocol was enforced, eliminating platelet count and derived mathematical ratios from the predictive feature space.

Experimental benchmarks on $N = 2032$ laboratory-confirmed NS1-positive dengue patients from Bangladesh proved that age-stratified modeling substantially outperforms conventional age-pooled baselines.

Explainable AI analysis using localized ‘shap.TreeExplainer’ attributions revealed that while pediatric and adult cohorts share strong macro-level feature importance agreement (Kendall’s τ -b = 0.900), the underlying non-linear decision thresholds diverge significantly by age. Fitting programmatic depth-1 decision trees to positive SHAP spaces uncovered distinct split cutoffs:

- **Lymphocyte Activation:** Split at $> 44.50\%$ in children (reflecting intense reactive lymphocytosis) vs $> 24.99\%$ in adults.
- **Hepatic Injury Surge:** Risk attributions for AST surge precipitously at ≈ 50 U/L in children, compared to a progressive slope beyond 80 U/L in adults.
- **Hemoconcentration Dominance:** Adult severe disease is heavily driven by elevated Hematocrit (split at $> 40.65\%$, ranking 3rd in global SHAP).

Independent external validation on a separate clinical cohort ($N = 50$) demonstrated robust generalizability (AUC 0.980, Accuracy 96.0%, MCC 0.917), confirming the trans-

ferability of the learned feature representations. Finally, the translation of these algorithms into a deployed React 19 + FastAPI clinical decision support web application bridges the gap between machine learning theory and clinical utility.

6.2 Summary of Key Contributions

1. **Age-Stratified Machine Learning Paradigm:** Demonstrated empirically that partitioning clinical dengue datasets by age significantly elevates diagnostic accuracy and sensitivity over age-pooled models.
2. **Zero Target Leakage Guarantee:** Formulated and verified a rigorous mathematical feature exclusion protocol isolating platelet counts from predictor feature sets.
3. **Subgroup-Conditioned Explainability & Threshold Discovery:** Extracted programmatic, clinically interpretable biomarker decision splits using shallow decision trees on SHAP attribution values.
4. **Independent External Generalizability:** Validated the zero-leakage model on an independent external clinical cohort, achieving 96.0% accuracy and 0.980 AUC.
5. **Production Full-Stack Clinical System:** Designed, implemented, and deployed a modern clinical dashboard featuring real-time SHAP waterfall attributions, dual-cohort comparison modes, and longitudinal snapshot tracking.

6.3 Future Work and Recommendations

Building upon the empirical findings and methodological boundaries established in this thesis, future research should pursue four high-impact directions:

1. **Prospective Multi-Center International Clinical Trials:** Deploy and evaluate the age-stratified model across hospital networks in South Asia, Southeast Asia, and Latin America using prospective patient cohorts with formally adjudicated WHO 2009 severity endpoints (Dengue with Warning Signs and Severe Dengue).
2. **Sequential Deep Temporal Modeling:** Utilize the longitudinal data storage infrastructure of our clinical dashboard to accumulate prospective multi-day complete blood count records and train Recurrent Neural Networks (LSTM, GRU) or Temporal Fusion Transformers for dynamic day-by-day disease trajectory forecasting.

3. **Multi-Omics & Biomarker Integration:** Incorporate rapid viral load quantification (RT-PCR cycle thresholds), inflammatory cytokine panels (IL-6, TNF- α , Ferritin), and serum albumin measurements to enhance early predictive sensitivity during the first 48 hours of illness.
4. **Edge Optimization for Point-of-Care Mobile Deployment:** Compress and quantize the ensemble models into ONNX / TensorFlow Lite runtimes integrated into offline-capable mobile applications for rural and resource-constrained clinics lacking continuous internet connectivity.

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